Project: Worked examples
Title: Weld group (elastic line method)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

A group of straight fillet welds in a plane under in-plane forces (with eccentricity, so torsion), an out-of-plane force and out-of-plane moments: the resultant force per unit length at every weld end, the critical point, the throat stress against the design strength and the leg size required.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Welds as lines of unit throat | $L_w = \Sigma\, l$, centroid $x_c = \Sigma\, l \cdot x_m / L_w$ |
| Unit second moments | $I_{ux} = \Sigma\,[l(y_m - y_c)^2 + l \cdot \Delta y^2/12]$, $I_{uy}$ likewise, product $I_{uxy} = \Sigma\,[l(x_m - x_c)(y_m - y_c) + l \cdot \Delta x \cdot \Delta y/12]$, $J_u = I_{ux} + I_{uy}$ |
| Torsion about the centroid | $T = F_y(e_x - x_c) - F_x(e_y - y_c) + M_z$ |
| At each weld end | direct $F/L_w$, torsional $T \cdot r/J_u$ (perpendicular to r), out-of-plane $F_z/L_w + [(M_x \cdot I_{uy} - M_y \cdot I_{uxy}) \cdot y' + (M_y \cdot I_{ux} - M_x \cdot I_{uxy}) \cdot x']/(I_{ux} \cdot I_{uy} - I_{uxy}^2)$ with $F_z$'s eccentricity added to $M_x$ and $M_y$ (this reduces to $M_x \cdot y'/I_{ux} + M_y \cdot x'/I_{uy}$ only for a group symmetric about a centroidal axis); vector sum $f_r$ |
| Throat and weld stress | $a = 0.707 \cdot s$, $\tau = f_{r,max}/a$; required leg $s_{req} = f_{r,max}/(0.707 \cdot f_w)$ |

Basis: Welds treated as lines: torsional and bending properties[1](11th ed., Ch. 9, Tables 9-1 and 9-2); Treating the weld as a line[2](1966, §7.4); Simplified method for design resistance of fillet welds (resistance input)[3](cl. 4.5.3.3).

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| welds | Weld lines | 3 row(s) | | drawing: BRK-010 rev A | any |
| s | Fillet leg | 8 | mm | drawing: BRK-010 rev A | 3 … 25 mm |
| $f_w$ | Design strength of the throat | 220 | MPa | assumption: S275, EN 1993-1-8 simplified method: 410/√3/(0.85 × 1.25) ≈ 223 MPa, rounded down. | 50 … 600 MPa |
| $F_x$ | In-plane force, x | 0 | kN | assumption: No horizontal load. | -100000 … 100000 kN |
| $F_y$ | In-plane force, y | -30 | kN | spec: Hoist rating with dynamic factor | -100000 … 100000 kN |
| $F_z$ | Out-of-plane force (tension positive) | 0 | kN | assumption: No out-of-plane load. | -100000 … 100000 kN |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| $e_x$ | Point of application, x | 250 | mm | drawing: BRK-010 rev A | -100000 … 100000 mm |
| $e_y$ | Point of application, y | 100 | mm | drawing: BRK-010 rev A | -100000 … 100000 mm |
| $M_x$ | Out-of-plane moment about x (tension on +y side) | 0 | kN·m | assumption: None. | -100000 … 100000 kN·m |
| $M_y$ | Out-of-plane moment about y (tension on +x side) | 0 | kN·m | assumption: None. | -100000 … 100000 kN·m |
| $M_z$ | Additional in-plane torque | 0 | kN·m | assumption: None. | -100000 … 100000 kN·m |

| TABLE ROW | VALUES |
|---|---|
| welds[1] | x1 = 0 mm; y1 = 0 mm; x2 = 80 mm; y2 = 0 mm |
| welds[2] | x1 = 0 mm; y1 = 0 mm; x2 = 0 mm; y2 = 200 mm |
| welds[3] | x1 = 0 mm; y1 = 200 mm; x2 = 80 mm; y2 = 200 mm |

## 3. Assumptions

A1. Elastic, static; the connected parts are rigid and the weld stress varies linearly with distance from the centroid. (Conservative against the instantaneous-centre method.)
A2. Fillet welds only, all of one leg size; the throat is 0.707·s. (Equal-leg fillet.)
A3. The design strength of the throat comes from S4 (EN 1993-1-8 simplified method) or a user allowable with its source. (No resistance table ships.)
A4. $f_w$ (Design strength of the throat) is taken as 220 MPa. (S275, EN 1993-1-8 simplified method: 410/√3/(0.85 × 1.25) ≈ 223 MPa, rounded down.)
A5. $F_x$ (In-plane force, x) is taken as 0 kN. (No horizontal load.)
A6. $F_z$ (Out-of-plane force (tension positive)) is taken as 0 kN. (No out-of-plane load.)
A7. $M_x$ (Out-of-plane moment about x (tension on +y side)) is taken as 0 kN*m. (None.)
A8. $M_y$ (Out-of-plane moment about y (tension on +x side)) is taken as 0 kN*m. (None.)
A9. $M_z$ (Additional in-plane torque) is taken as 0 kN*m. (None.)

## 4. Working

[11] $L_w = \Sigma\ l$
$= 80 + 200 + 80$
$= 360\ mm$
Total weld length

[12] $x_c = \Sigma\ l \cdot x_m\ /\ L_w$
$= 17.778\ mm$
$= 17.778\ mm$
Centroid x

[13] $y_c = \Sigma\ l \cdot y_m\ /\ L_w$
$= 100\ mm$
$= 100\ mm$
Centroid y

[14] $I_{ux} = \Sigma\ [l(y_m - y_c)^2 + l \cdot \Delta y^2/12]$ [1](Table 9-2)
$=$ about the centroid
$= 2266700\ mm^3$
Unit second moment about x

[15] $I_{uy} = \Sigma\ [1(x_m - x_c)^2 + 1 \cdot \Delta x^2/12]$ [1](Table 9-2)
= about the centroid
= 227560 mm³
Unit second moment about y
[16] $I_{uxy} = \Sigma\ [1(x_m - x_c)(y_m - y_c) + 1 \cdot \Delta x \cdot \Delta y/12]$ [1](Table 9-2)
= about the centroid
= 0 mm³
Unit product moment
[17] $J_u = I_{ux} + I_{uy}$ [1](Table 9-1)
= 2266700 + 227560
= 2494200 mm³
Unit polar moment
[18] $T = F_y(e_x - x_c) - F_x(e_y - y_c) + M_z$
= -30 × 232.22 − 0 × 0 + 0e3
= -6.9667 kN·m
In-plane torque about the centroid
[19] $f_{r,max}$ = max over weld ends $\sqrt{(f_x^2 + f_y^2 + f_z^2)}$
= at (80, 0) mm: (-279.31, -257.13, 0) N/mm
= 379.64 N/mm
Maximum resultant force per unit length
[20] critical = critical point
= (80, 0) mm
= "(80, 0) mm"
Where $f_r$ is greatest
[21] a = 0.707 * s
= 0.707 × 8 mm
= 5.656 mm
Throat
[22] $\tau = f_{r,max} / a$
= 379.64 N/mm / 5.656 mm
= 67.122 MPa
Maximum throat stress
[23] $s_{req} = f_{r,max} / (0.707 * f_w)$
= 379.64 N/mm / (0.707 × 220 MPa)
= 2.4408 mm
Leg size required
[24] $u = \tau / f_w$
= 67.122 MPa / 220 MPa
= 0.3051
Utilisation

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $L_w$ | 360 mm | Weld length |
| $x_c$ | 17.778 mm | Centroid x |
| $y_c$ | 100 mm | Centroid y |
| $I_{ux}$ | 2266700 mm³ | Unit second moment about x |
| $I_{uy}$ | 227560 mm³ | Unit second moment about y |
| $I_{uxy}$ | 0 mm³ | Unit product moment |
| $J_u$ | 2494200 mm³ | Unit polar moment |

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| T | -6.9667 kN·m | Torque about the centroid |
| $f_{r,max}$ | 379.64 N/mm | Maximum force per unit length |
| critical | "(80, 0) mm" | Critical point |
| a | 5.656 mm | Throat |
| τ | 67.122 MPa | Throat stress |
| $s_{req}$ | 2.4408 mm | Leg required |
| u | 0.3051 | Utilisation |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Throat stress against design strength | u = 0.3051 | u ≤ 1 | 0.305 | [3](cl. 4.5.3.3) |

## 7. Applicability

No warnings.

## 8. Verification

Template mech.weld-group@1.1.1 Content hash d7b4436f57b87af7 Engine calc-engine/1.0.0
Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

## 9. References

1. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.
2. Blodgett, Design of Welded Structures.
3. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-8: Design of Joints. BS EN 1993-1-8:2005. BSI; 2005.